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B.Tech. Degree VII Semester Examination November 2016

IT 1705 (E2) CRYPTOGRAPHY AND DATA SECURITY (2012 Scheme)

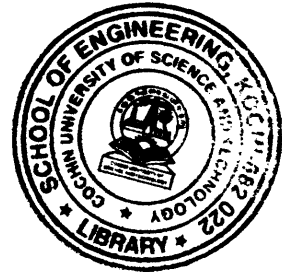
Time: 3 Hours

Maximum Marks: 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define cryptography and cryptanalysis.
- (b) Write a note on unicity distance.
- (c) What is stream and block enciphering?
- (d) Explain the Golomb's three criteria for random sequence.
- (e) Explain knapsack system.
- (f) Define zero knowledge technique.
- (g) Explain fair cryptosystems.
- (h) Explain Diffie-Hellmann protocol.



PART B

(4 × 15 = 60)

- II. Explain the classical cipher system in detail. (15)
- OR**
- III. Explain the working of Hagelin machine. (15)
- OR**
- IV. (a) Explain IDEA. (7)
- (b) Explain the mode of DES. (8)
- OR**
- V. Explain DES algorithm in detail. What are the characteristics of DES? (15)
- OR**
- VI. Explain RSA algorithm in detail. (15)
- OR**
- VII. Explain the message authentication with digital signature. (15)
- OR**
- VIII. (a) Explain the general aspects of key management (8)
- (b) Explain network security. (7)
- OR**
- IX. Explain key distribution of asymmetrical systems. (15)